Figure 3. Local edge pulses expose coupled boundary motion and test an untuned prediction of polarity-domain drift.

A. Native Wave-pinning active species u versus arc position centred on the incumbent's initial centre, at C = 30 for (i) gap −1 (straddling) and (ii) gap 4 (separated). Rows show 0%, 50% and 90% of the target heading under continuous cue, then the final cue-free profile. Grey shading marks the fixed cue support in every row; open diamonds mark an inter-region saddle where resolved, and black ticks on the arc axis mark heading. Each column shares its y-scale.

B. Time to 50% of the angle from initial activity-vector heading to the cue direction versus initial cue edge gap (near edge of cue support minus the incumbent's 10%-contrast boundary).. Time and length retain each model's native units. Curves, gaps, brackets and three Figure 3 sizes are unchanged; lighter solid, intermediate dashed and darker dotted lines denote increasing C. Filled i–ii markers identify A.

Vertical lines mark the first tested gap with two resolved regions and a saddle at the 50% crossing for each C, using that size's line style; the existing periodic watershed detector is applied to the native saved-field interpolant. Light grey bands mark sampled near-antipode placements with undefined heading at one or more sizes (activity-vector coherence $R < 0.05$ or a cancellation interval); that size's curve breaks there. Open up-triangles mark censored time lower bounds; horizon caps are not measured values.

For the relaxation row only, cue-off occurs at 95% heading or at 20,000 if 95% is not reached, restarting from the full native event state after an identical cue-on replay. No-cue relaxation stops when max $|\partial u/\partial t|$ reaches the existing tolerance 2.2683×10^{-7} (native activity per native time; the original 10^{-4} scaled-RHS tolerance over 1,000 time units), or after four times the cue-on duration. i: cue off at 95% heading, t = 790.5; relaxed for 258.6, stopped at stationarity; ii: cue off at 95% heading, t = 227.6; relaxed for 291.3, stopped at stationarity.

C. Control-subtracted displacement of the pulsed falling half-contrast edge, divided by peak input a, at f = 0.001. Shading denotes Figure 4 circumference: chemical C = 20, 30, 40, 50, 60; Spring resting C = 1, 1.5, 2, 2.5, 3. The stationary polarized state at each declared constant mean is used, without noise or retuning. A Gaussian pulse centered on the boundary has spatial SD $\ell = 1.5$ and duration $\tau = 50$ (Spring 0.15 and 5). Peak a is 10^{-4} of the native reference: 10^{-5} Wave-pinning, 10^{-4} Holmes, 4×10^{-4} Spring and 6.4×10^{-4} Goryachev. The grey interval is pulse-on. Inset $A_{20}\tau$ is displacement integrated through 20τ divided by pulse dose $a\tau$. Both edges, a/2 and 2a are checked in the supplement. A residual positional offset is allowed because translation is neutral; this is not an infinite displacement integral. Largest/smallest terminal displacement per pulse dose is 0.701 Wave-pinning, 0.914 Holmes, 0.989 Spring and 0.0878 Goryachev.

D. The prescribed independent-edge prediction, $D_s = (D_{left} + D_{right})/4$, against physical tracked domain drift measured in Figure 4 at $f \leq 0.01$ (markers). Colors and shading match C. Each measured terminal edge gain per pulse dose scales the native spatial response shape, normalized to give unit response to that exact localized pulse; this extra spatial-shape assumption is explicit. The native linearization supplies its shape only, while measured C

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amplitudes supply its gain; the imposed discrete spatial covariance and phase-averaged OU-knot interpolation enter the exact lognormal covariance $\mu^2[\exp(\sigma^2 C_x C_t) - 1]$, with $\sigma = 1$. There is no diffusion fitting or parameter adjustment. Line: 1:1; horizontal bars: Figure 4 whole-seed-bootstrap 95% CI. This predicts Figure 4's tracked drift, not total heading decay. The independent prediction/observation medians are Wave-pinning 0.511, Holmes 0.540, Spring 0.534, Goryachev 0.437. Goryachev spans 0.00359–4.01 across conditions, so a single bias does not summarize its failure. Goryachev C20/C30 at f = 0.01 has no polarized boundary at the declared mean, so its prediction is undefined.

The edge-independence supplement measures simultaneous half-contrast displacements from existing Figure 4 exemplar profiles, seeds 0 and 1, using uninterrupted single-region intervals and exposing float16 precision limits. At quarter-window lag, resolved median correlations are Wave-pinning 0.980, Holmes 0.983, Spring 0.997, Goryachev 0.759. Positive common motion violates edge independence. Retaining the native shared translation covariance doubles the prediction; this sensitivity is supplementary and does not adjust any parameter. Goryachev's disagreement and short valid-region exposure remain explicit.

Cellular counterpart of C: photoactivation straddling a leading-edge boundary, tracking both flanking boundaries and their displacement after withdrawal. Native units are not calibrated between models. deBelly is excluded from rings.

$$D_s^{\mathrm{ind}} = \frac{D_L + D_R}{4}, \qquad D_s = \frac{D_L + D_R + 2D_{LR}}{4} \tag{3.1}$$

The main panel applies edge independence; the supplement retains the measured positive boundary coupling.

$$\widehat{g}_{e,j} = \frac{M_e}{Q_e} g_j, \qquad M_e = \frac{\Delta x_e(20\tau)}{a\tau}, \quad Q_e = \sum_j g_j h_{e,j} \tag{3.2}$$

Measured terminal displacement per pulse dose supplies gain. The unchanged native linearization supplies only the spatial shape g; h is the actual localized pulse shape. No measured diffusion enters.

$$K_u(j,k;t,t+v) = \mu^2\{\exp[\sigma^2 C_x(j-k) C_t(t,t+v)] - 1\} \tag{3.3}$$

This is the declared lognormal covariance, with the exact discrete circular spatial kernel and variance-normalized OU-knot interpolation; the knot phase is averaged.

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$$D_e = \sum_{j,k} \widehat{g}_{e,j} \widehat{g}_{e,k} \int_0^\infty \overline{K_u(j,k;t,t+v)}^{t} \, \mathrm{d}v \tag{3.4}$$

Native linear response and the convention MSD = 2Dt give each marginal edge diffusivity. The time integral is evaluated without generating new stochastic inputs.